DO NOW — RECALL

Stacks and queues

Name: _____ Class: _____ Date: _____

1. What is a[1] for int[] a={6,9,3}?

2. What is −1 in 8-bit two's complement?

3. A loop running from 0 to 3 inclusive executes how many times?

4. Push 2, push 5, pop once: which value is removed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

9.

2. Model answer

11111111.

3. Model answer

4.

4. Model answer

5, the last item pushed.

Vocabulary — know these terms

stack — LIFO abstract data type.

push — Add an item onto a stack.

pop — Remove the top stack item.

queue — FIFO abstract data type.

dequeue — Remove the front queue item.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm and explain

Remove A: front=1, count=2. Insert D at slot0: rear=1, count=3. Physical D,B,C differs from logical B,C,D; modulo reuses the freed slot.

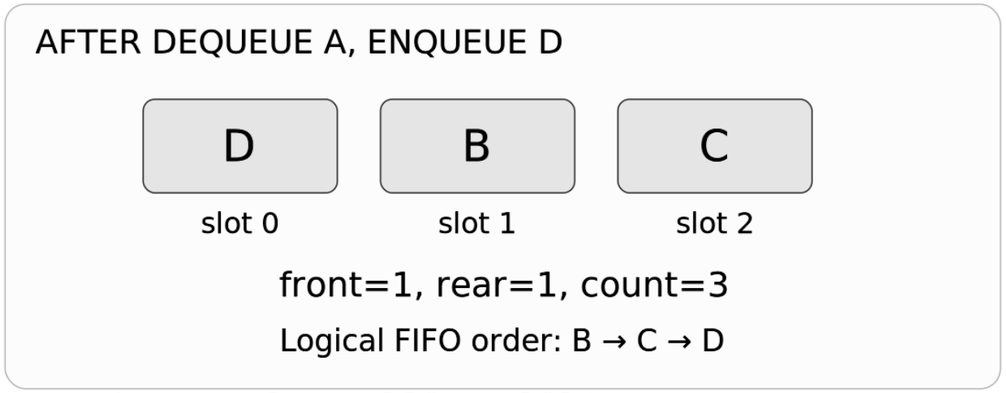
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a circular queue of capacity 3.

After A,B,C: front=0, rear=0, count=3. The queue is full even though front==rear. rear means next free slot here, unlike Lesson 020.



After dequeue A and enqueue D: physical D B C; logical B C D; front one rear one count three

NEW LEARNING

Read, annotate and explain

Push

Adding an item puts it on top of a stack.

Pop

Removing an item takes the most recently pushed item: 7.

Contrast with a queue

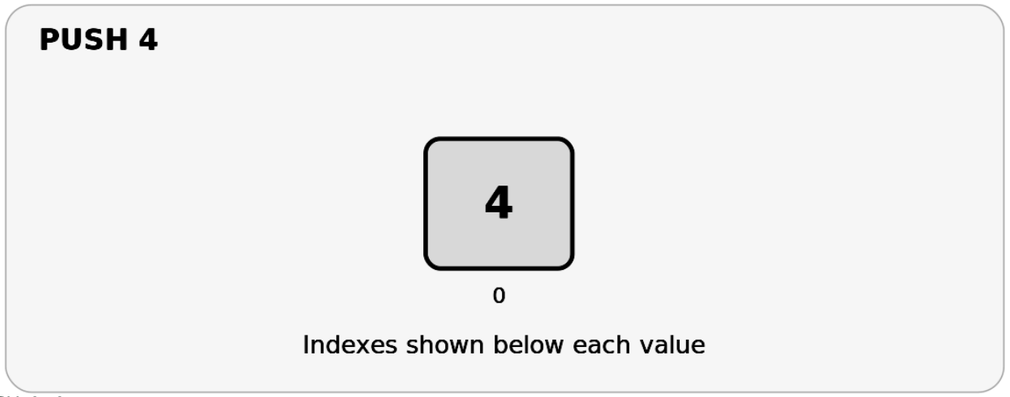
A queue removes the oldest item instead (FIFO); it is suited to arrival-order processing.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

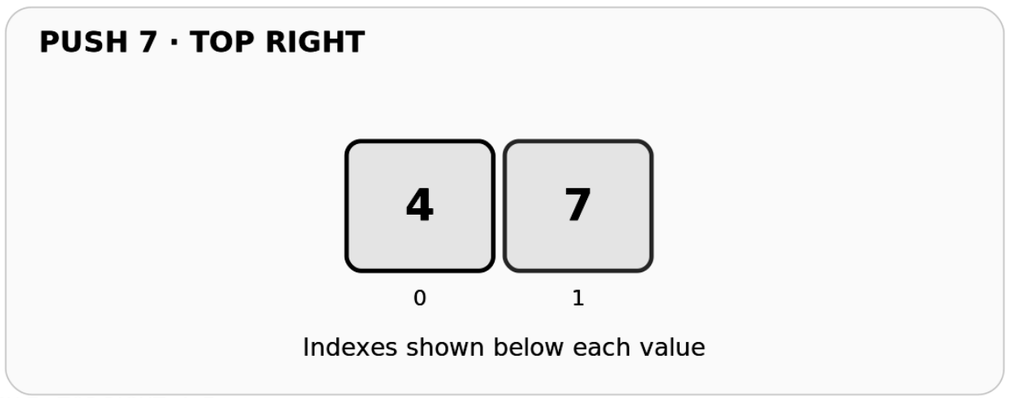
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



PUSH 4. 4

New Learning



PUSH 7 · TOP RIGHT. 4, 7

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A queue removes the most recently inserted item first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

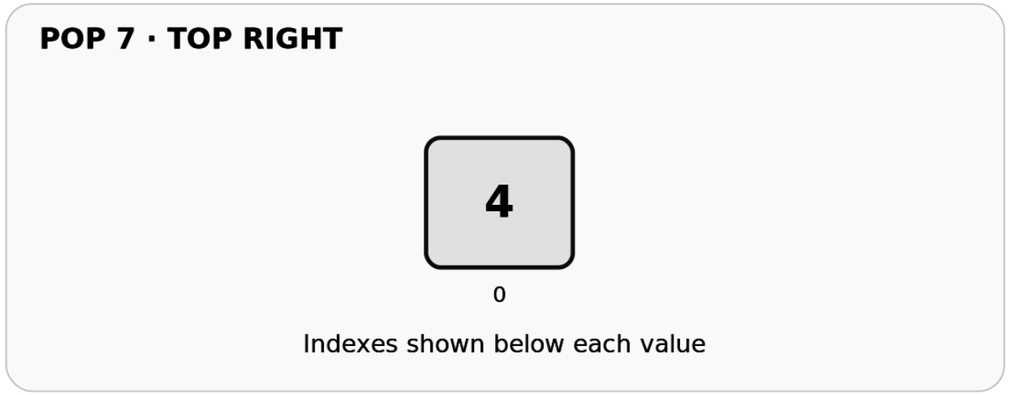
Question 3

Explain why an undo history usually uses a stack rather than a queue.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



POP 7 · TOP RIGHT. 4

I DO — WORKED EXAMPLE

Follow the worked method

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

Predict

The newest value 5 is on top, so Pop removes 5.

Confirm and explain

The stack remains [3,8] from bottom to top.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Push A, B, C onto a stack then Pop twice. What remains?

Question 2

A circular array queue has capacity 3. Starting empty, enqueue 10,20,30; dequeue once; enqueue 40. State physical slots, front/rear/count and logical order. Use rear as the next free slot.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict After A,B,C: front=0, rear=0, count=3. The queue is full even though front==rear. rear means next free slot here, unlike Lesson 020.

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

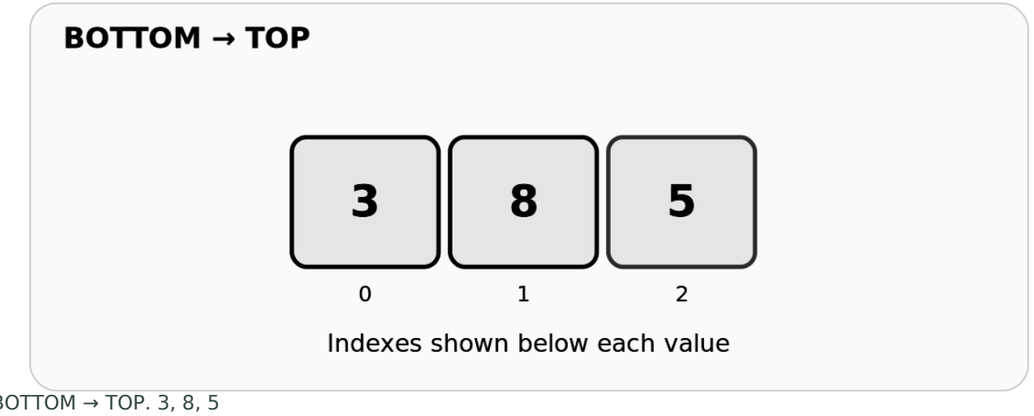
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

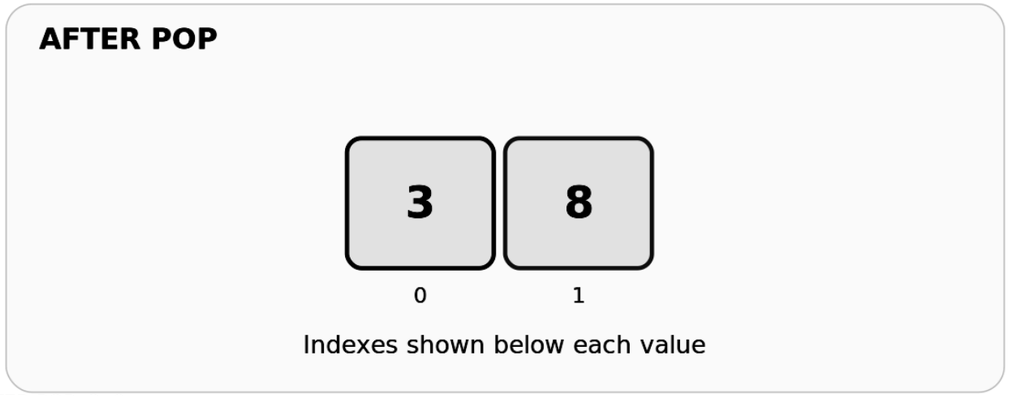
The newest value 5 is on top, so Pop removes 5.



BOTTOM → TOP. 3, 8, 5

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

The newest value 5 is on top, so Pop removes 5.



AFTER POP. 3, 8

WE DO — GUIDED PRACTICE

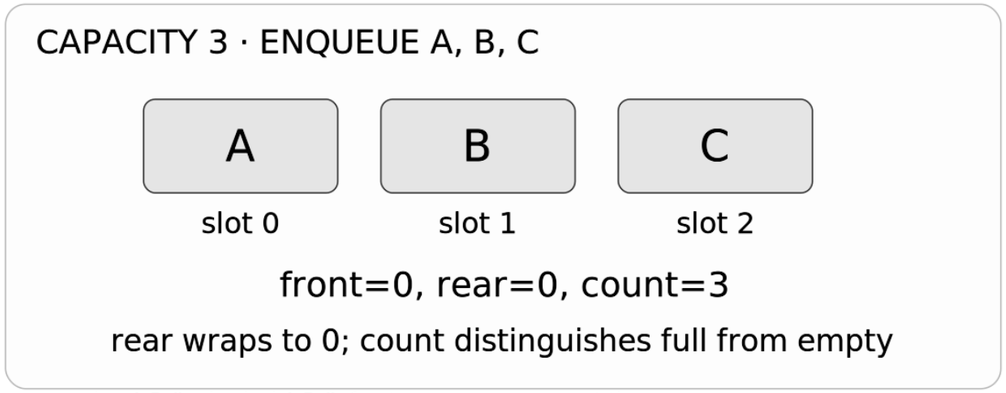
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a circular queue of capacity 3.

After A,B,C: front=0, rear=0, count=3. The queue is full even though front==rear. rear means next free slot here, unlike Lesson 020.



After enqueuing A B C: physical A B C; front zero rear zero count three